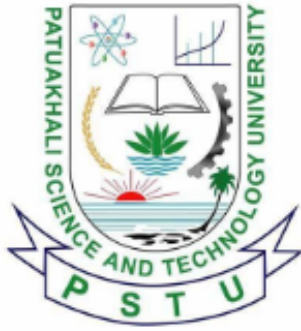




Patuakhali Science and Technology University

Faculty of Computer Science and Engineering

CCE 322 :: Computer Peripheral and Interfacing Sessional

Final Report



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Farm Droid

An Autonomous Agricultural Robot

1. Introduction

Labour shortage and the requirement to monitor crop health with precision are huge problems in agriculture. To tackle these challenges, it has developed a novel agricultural robot concept known as “Farm Droid”. To overcome these challenges, it has come up with a novel idea for an agricultural robot, dubbed “Farm Droid”, which is compact and autonomous.

Farm Droid is the low-cost, adaptable mobile platform that can be used in place of traditional heavy machinery. This enables it to maneuver between rows of crops efficiently to carry out monitoring and manipulating. It combines cutting-edge embedded technologies: a Raspberry Pi 4 computer for computer vision and decision making, and an arduino UNO/ nano board for sensor data collection and real time motor control.

2. Objectives

The main goal is to create a smart farming assistant that will be able to work autonomously. The aims are:

1. To design and develop a lightweight and low cost agricultural robot that can move between the rows of crops.
2. To implement a multi-sensor system to monitor important environmental and crop health parameters continuously such as soil moisture, soil temperature, rainfall, and soil pH.
3. To build a distributed computing system with high level vision processing and client application for real time monitoring and control.
4. To build weeding tool, soil collector etc as modular attachments of the Robotic arm.
5. To support auto digging and seeding operations in a small scale environment.

3. Problem Statement

Traditional large-scale agricultural equipment may not be suitable for small-scale precision agriculture or sensitive crop fields. Large tractors compact soil do not have the agility to do individual plant inspection. The robotic systems available today are either expensive or complex enough to be too large to operate in these environments. This is where Farm Droid comes in with its lightweight, intelligent and versatile solution.

4. Scope

The project scope also includes the mechanical assembly of the robot and programming of the circuit, as well as the integration of the circuit into the robot.

The hardware design consists of a modified chassis, motor / servo drive system, and a power distribution system to insure stable operation of the hardware components. The apparatus is basically made of a camera and a motor. The software side of the system includes OpenCV vision processing using Python and motor control based on a C++ firmware.

The challenge with this project is that this prototype won't be deployed at a large scale industrial application level; it will only be spun in a small scale environment such as a garden or a test bed under control.

5. Methodology

5.1. System Architecture

The computational load of the system is efficiently dealt with with a “Distributed Computing” approach. It is an architecture that includes the Brain (Raspberry Pi 4), responsible on the high level for the decision making, image processing (OpenCV) and path planning. It also communicates with other modules with interfaces such as Serial, I2C and USB. The Controller (Arduino Uno / Nano) is used for low level real time control, for instance sending PWM impulses to motor control drivers or reading analog sensor data.

5.2. Technology Stack

1. **Language:** Python (Vision/Logic), C++ (Arduino Firmware)
2. **Accelerator:** Intel® RealSense™ depth camera (Optional for object detection)
3. **Hardware Interface:** I2C (Inter-Integrated Circuit), USB.
4. **Power Management:** 65W/22W 30000mAh (Buck Converters) for system stability.

6. Hardware Components

The Farm Droid consists of the following elements:

1. **Actuators:** DC Gear Motors or MG996R Servos (Mobility), SG90 (Gripper/Arm), GA25 (Metal Gear motor).
2. **Sensors:** Ultrasonic (HC-SR04), Soil Moisture, Rain Sensor, DHT11, IR (Infrared), LDR (Light Dependent Resistor), DFPlayer Mini, RFID Reader(Radio Frequency Identification), Relay, PH Electrode, Gas sensor(MQ135).
3. **Controllers:** Raspberry Pi (4 model B (4Gb)), arduino Uno / Nano.
4. **Drivers:** L298N/Motor Driver (for Wheels).
5. **Vision:** Official Raspberry Pi Camera Module with Pi camera 5MP.
6. **Power Supplier:** LM2596 Buck Converter, 65W/22W 30000mAh Power Bank.

7. Budget Estimation

The costs of the various components that will be needed for the “Farm Droid” are summarized below and based on actual market prices in Bangladesh. .

Category	Component Name	Unit (BDT)	Qty	Total (BDT)
Controllers	Raspberry Pi 4 Model B (4GB) [6]	14,100	1	14,100
	Arduino Uno R3 [4]	988	2	1,976
Motors & Drivers	GA25 Metal Gear Motor [3]	450	2	900
	DSServo DS3218 20KG Digital Metal Servo [4]	2350	2	4700
	Servo Motor MG996R [6]	500	1	500
	Servo Motor SG90 (Micro) [4]	180	2	360
Sensors	HC-SR04 Ultrasonic Sensor [2]	99	1	99
	FC-51 IR Obstacle Sensor [2]	45	3	135
	HC-SR501 PIR Motion Sensor [4]	93	1	93
	OV7670 Camera Module [2]	400	1	400
	DHT11 [3]	95	1	95
	LDR (Light Dependent Resistor) [3]	5	1	5
	DFplayer [4]	350	1	350
	PIR Sensor [4]	93	1	93
	RFID [4]	190	1	190
	Relay [4]	80	4	320
	PH Sensor [4]	2584	1	2584
	Moisturizer Sensor [4]	70	1	70
	Rain Sensor [4]	78	1	78
	MQ135 [4]	135	1	135
Power & Audio	Breadboard Power Supply [4]	76	1	76
	3 Watt 8 Ohm Mini Speaker [4]	250	1	250
	Electret Microphone [6]	15	2	30
Mechanical & Misc	PVC Pipe	300	5	1,500
	Pipe (Small)	30	5	150
	Half-Size Breadboard [2]	75	2	150
	Jumper Wires (Male-Male) [2]	100	4	400
	USB Cable (Arduino) [4]	150	2	300
	Screws / Hardware	100	10	1,000
	Copper Wire (26 SWG) [2]	110	2	220
Tools	Digital Scale (10kg) [1]	259	1	259
	Hex Saw	15	2	30
	Sand Paper	30	3	90
Grand Total Estimated Cost				31488

Table 1: Detailed Budget Estimation

8. Analysis of cost-benefit

This section explains the cost-benefit analysis of the “Farm Droid” from a financial feasibility/practical viability viewpoint. This analysis contrasts the initial development and operational costs against the tangible and intangible benefits generated over a projected 3-year operational lifecycle.

8.1. Cost breakdown

The costs can be broken down to initial capital spending (CapEx) and regular operating spending (OpEx).

Cost Category	Amount (BDT)	Amount (USD)
Capital Expenditure (CapEx)	33,000	270
Operational Expenditure (OpEx - Annual)	3,000	25
Total anticipated cost over three Years		42,000

Table 2: Cost breakdown and total anticipated cost over three Years

8.2. Benefit analysis

Benefit Type	Description
Tangible Benefits	Automated labor savings, yield protection through early detection, and resource efficiency lead to an estimated annual financial benefit of BDT 27,000.
Intangible Benefits	Soil preservation through reduced compaction, data-driven insights for better crop management, and improved safety for farm workers.

Table 3: Summary of Tangible and Intangible Benefits

8.3. Market Comparison: Farm Droid vs. Commercial Alternatives

The overall estimated budget is about 258 USD (with an exchange rate of 122 BDT/USD). Comparing with commercial robots, we get the following insights:

Robot Model	Brand	Primary Function	Price (USD)
Farm Droid (Our Prototype)	Custom	Precision Monitoring & Manipulation	258
ATEAGO 50W [7]	ATEAGO	Remote Lawn & Terrain Maintenance	4,798
ATEAGO 50PRO [8]	ATEAGO	Remote Lawn & Terrain Maintenance	7,199
ATEAGO 80PRO [9]	ATEAGO	Heavy Terrain Maintenance	14,999
ADIR Basic [10]	Ant Robotics	Autonomous Ag-Logistics & Scouting	23,292
ADIR Power [11]	Ant Robotics	Heavy Autonomous Ag-Logistics	29,115

Table 4: Price Comparison with Commercial Agriculture Robots (Source: RobotShop)

9. Comparison with related works

The following table shows a comparison of the features of our prototype with some works as of now.

Robot Model	Primary Function(s)	Locomotion & Control	Key Features & Innovations
Farm Droid (Our Prototype)	Real-time monitoring, robotic manipulation, and weeding	Rover, Raspberry Pi 4 (brain), Arduino (controller)	Real time monitoring with weeding, modular robot arm, as well as opencv for image detection. Alongside Gemini is used for LLM integration.
Modular 4WD Robot [12]	Grass cutting, leaf crushing, and precision seeding	4WD skid-steer, LiDAR/Camera, EKF state estimation	Features a vertically stacked two-unit architecture with quick-release toolheads. Uses simulated coverage-path algorithms (like zigzag/boustrophedon) for 100% gap-free field coverage.
AgriCruiser [13]	Over-the-row precision spraying for weed management	2-wheel front differential drive with rear casters	Open-source, highly reconfigurable chassis made from T-slot extrusions. Offers adjustable track width (1.42–1.57 m) and high clearance at a low deployment cost of 5,000–6,000.
Six-Wheeled Laser Weeder [14]	High-precision laser weed elimination	6-wheeled, double four-bar linkage suspension	Uses a 3-axis linear actuation mechanism to position a 10W blue diode laser directly over weeds, achieving a 97% hit rate while navigating obstacles up to 15 cm.
FarmDroid FD20[15]	Autonomous seeding and preventative mechanical weeding	4-wheeled, solar-powered, GPS-guided	Operates on a memory-based approach, recording exact GPS coordinates

			of planted seeds to mechanically weed between and within rows without complex real-time vision. Heavily studied for climate-smart soil management.
BoniRob [16][17]	Crop monitoring, targeted fertilization, and weeding	4-wheel independently steered	Adjusts working width dynamically (0.75m–2.0m) and uses mechanical cylinders to crush weeds as an alternative to herbicide spraying.
ROBOTTI 150 D [18]	Multipurpose field operations (evaluated for soil health impact)	Lightweight wheeled vehicle	Used extensively in research to quantify soil compaction. Studies show it causes less sub-soil compaction than heavy tractors, though repeated wheeling still alters topsoil porosity.
Tertill [19]	Home garden weed control	Solar-powered mini rover	A consumer-grade, waterproof robot that autonomously navigates small plots and uses a simple wire/string trimmer to cut newly sprouted weeds.

Table 5: Feature Comparison with Commercial Agriculture Robots (Source: RobotShop)

10. Work Timeline

Task	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6
Requirement Analysis & Planning	✓					
Mechanical Assembly (Chassis & Drive)	✓	✓				
Circuit Integration & Power Distribution		✓	✓			

Sensor Installation & Wiring		✓	✓			
Motor Control & Calibration			✓	✓		
Arm Control & Actuation Testing				✓	✓	
Software Integration				✓	✓	✓
System Testing & Debugging					✓	✓
Final Review & Documentation						✓

Table 6: Gantt Chart for Farm Droid Development

11. Visual Models

11.1. Circuit Diagram

This diagram shows the circuit design alongside pin diagram of each components, for easier navigation and reproducibility.

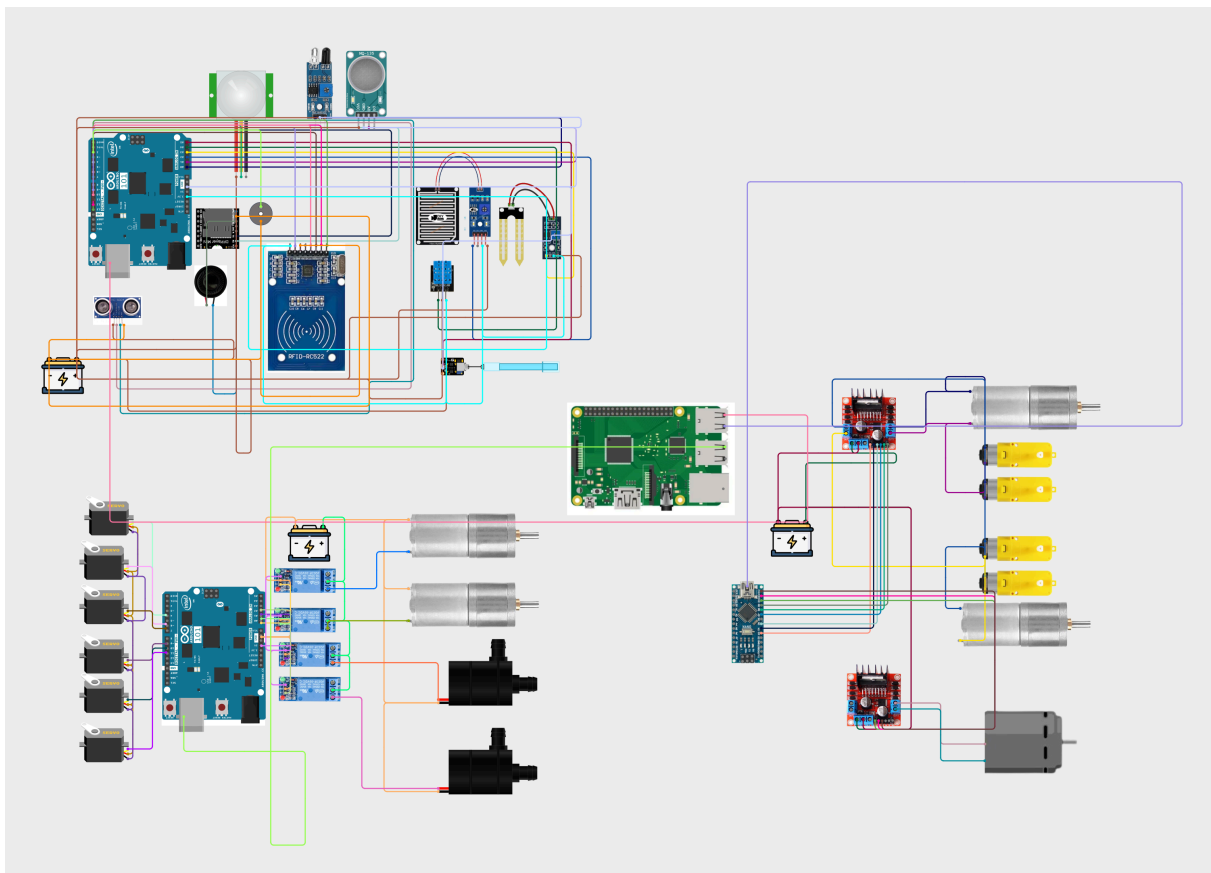


Figure 1: Circuit Diagram of Farm Droid System

11.2. Flow Chart

The flow chart shows the overall flow of the system, from the input of the sensors to the output of the actuators.

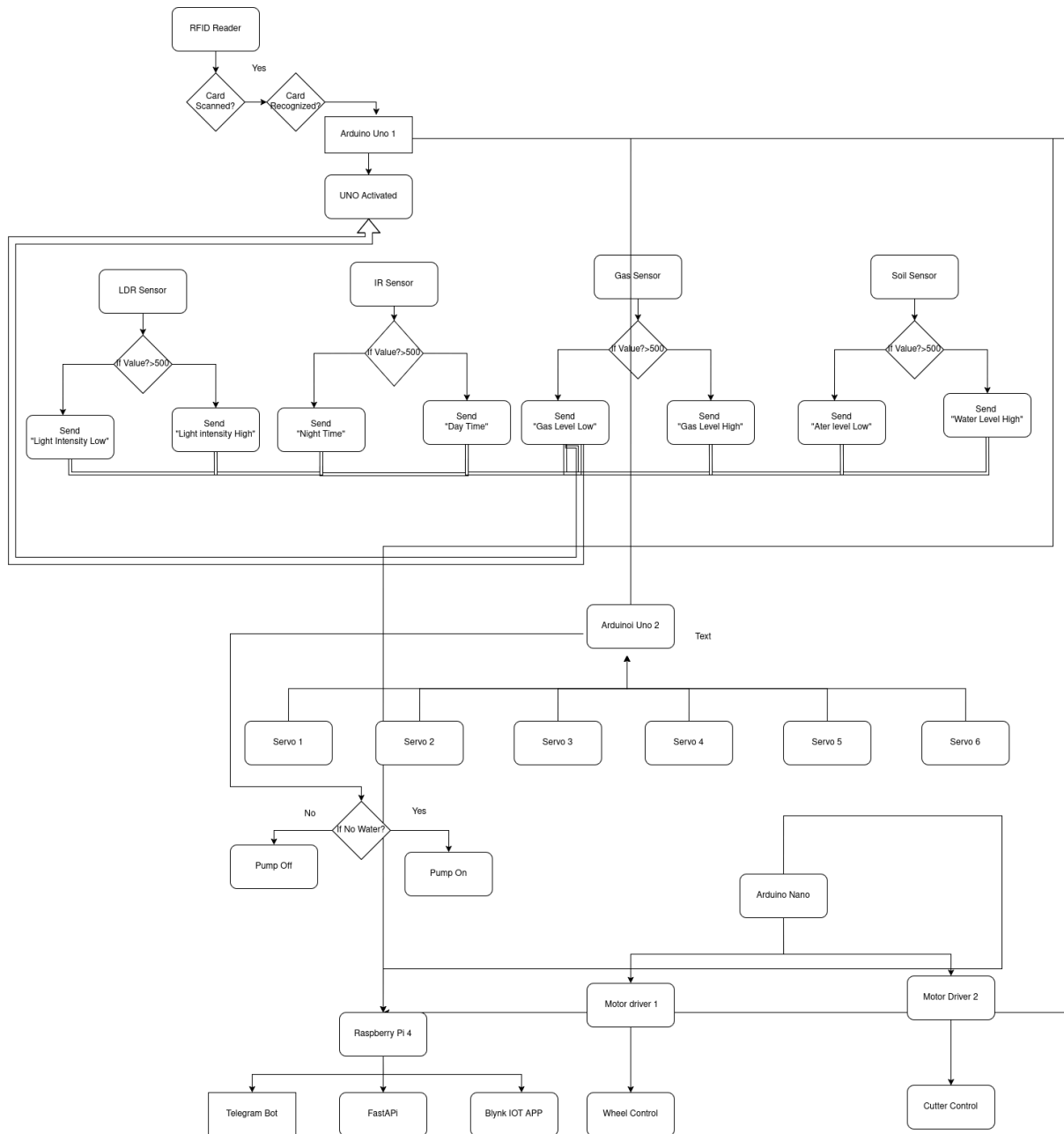


Figure 2: Flow chart of Farm Droid system

11.3. Data Flow Diagram

The data flow diagram (Fig 3) shows the flow of data between the different components of the system, including the sensors, controllers, and actuators.

12. Results

This project shows the successful development of a prototype. Our bot is able to carry out basic agricultural activities such as monitoring soil moisture, temperature, rainfall and soil pH. The integration of computer vision allows it to detect obstacles, as well as perform weeding operations. The modular design of the robotic arm enables it to be used for various tasks in the future. And finally, it can also dig the ground and plant seeds in a small scale environment.

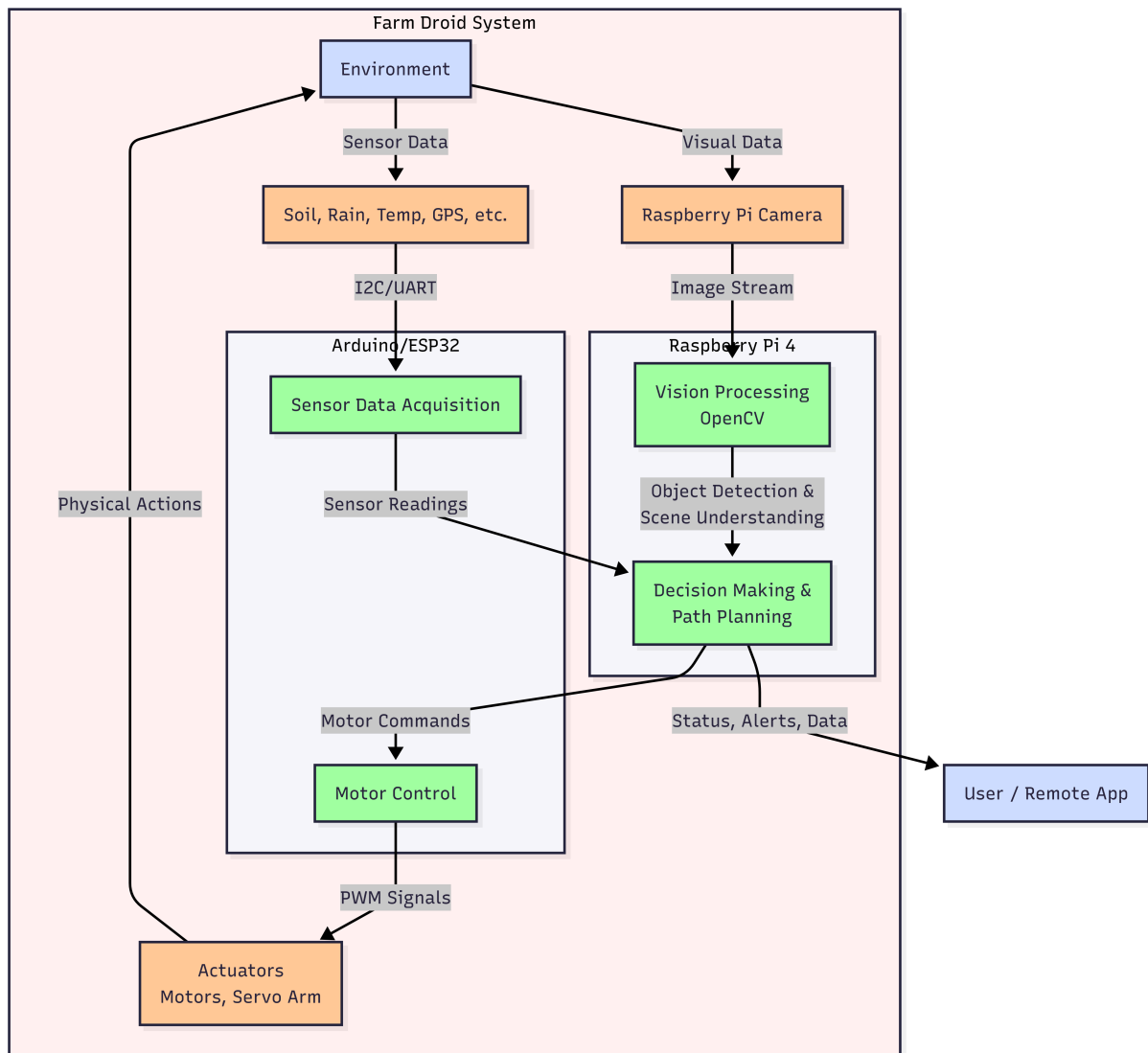


Figure 3: Data flow diagram of Farm Droid system

13. Future Plans

1. To design and develop weeding tool, seed dispenser, watering nozzle etc as modular attachments of the Robotic arm.
2. Due to limited battery capacity, our project can't run continuously for a long time.

14. Conclusion

The Farm Droid is a step towards modernising agriculture by robotics. Hopefully this project will explore the potential of autonomous mobility and computer vision, to prove farming automation can be fast, intelligent and accessible to all. The successful completion of this project will result in a working prototype that can be used to move autonomously along paths and carry out basic agricultural activities.

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